

LAB TASK # 7

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SAP ID # 70150006 [Company address]

.MODEL SMALL

.STACK 100H

.DATA

num1 db 127

num2 db 2

result db 20 dup('<\$>')

nuMber DW 128

VALUE DB 10110101B

COUNT DB 0

;VALUE DB 10101010B

.CODE

MAIN PROC

MOV AX,@DATA

MOV DS,AX

CALL PROC1

call proc2

call proc3

call proc4

```
MOV AH,4CH  
INT 21H  
MAIN ENDP  
;END MAIN
```

```
PROC1 PROC
```

```
    jc unsinged_overflow  
    jo signed_overflow  
    jmp done
```

```
unsinged_overflow:  
    mov result,0ffh  
    jmp done
```

```
signed_overflow:  
    mov result,7fh
```

```
done:  
    ret
```

```
PROC1 ENDP
```

.....TASK1
.....

; .data

; .CODE

PROC2 PROC

MOV AX,NUMBER

SHR AX,1

SHR AX,1

SHR AX,1

RET

PROC2 ENDP

.....TASK2
.....

; .DATA

; .CODE

PROC3 PROC

MOV AL,VALUE

MOV CL,8

MOV BL,0

COUNT_LOOP:

ROL AL,1

JNC SKIP_COUNT

INC BL

SKIP_COUNT:

DEC CL

JNZ COUNT_LOOP

MOV COUNT,BL

RET

PROC3 ENDP

;;;;;;;;;;;;;TASK3

PROC4 PROC

MOV AL,VALUE

STC

RCL AL,1

RCR AL,1

RET

;;;;;;;;;;;;;TASK4

PROC4 ENDP

END MAIN

